RAJALAKSHMI ENGINEERING COLLEGE

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N2

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|---|---|
| **Started on** | Thursday, 24 September 2026, 10:28 AM |
| **State** | Finished |
| **Completed on** | Thursday, 24 September 2026, 10:28 AM |
| **Time taken** | 31 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
int n,k,i=0,j=1;
scanf("%d",&n);
int a[n];
for(int x=0;x<n;x++)
scanf("%d",&a[x]);
scanf("%d",&k);
while(i<n&&j<n){
if(i==j){
j++;
}
else if(a[j]-a[i]==k){
printf("1");
return 0;
}
else if(a[j]-a[i]<k){
j++;
}
else{
i++;
}
}
printf("0");
return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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